

Class 35 Cantor's Diagonalization Proofs

Goals

1. Prove the rational numbers are countable.
2. Prove the real numbers are uncountable.
3. Know Cantor's Theorem

Useful Theorems

A countable union of countable sets is countable.

Any subset of a countable set is countable.

Any set that contains an uncountable set is uncountable.

$(1,4)$	$(2,4)$	$(3,4)$	$(4,4)$
$(1,3)$	$(2,3)$	$(3,3)$	$(4,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$
$(1,1)$	$(2,1)$	$(3,1)$	$(4,1)$

Define a bijection
between the elements
in $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$
by creating a list
of all the points

1 2 3 4 5 6 7 8 9 10

Let Q_0 be the set of rational numbers in the interval $[0,1)$.
Let Q_1 be rationals in $[1,2)$ and so forth for Q_2 , Q_3 ,

Create a list of elements in Q_0 to prove that the set Q_0 of rational numbers in $[0,1)$ is countable.

$(1,4)$	$(2,4)$	$(3,4)$	$(4,4)$
$(1,3)$	$(2,3)$	$(3,3)$	$(4,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$
$(1,1)$	$(2,1)$	$(3,1)$	$(4,1)$

Cantor's second diagonal method is the key to proving the set of real numbers is uncountable.

We will prove that the open interval $(0,1)$ is uncountable. This forces us to conclude the entire set of real numbers is uncountable since it contains an uncountable subset.

Assume for contradiction that $(0, 1)$ is a countable set.

Cantor's Theorem: If A is any set then, the cardinality of A is strictly less than the cardinality of the power set of A .

Proof:

Define a function f with domain A and codomain $P(A)$ as follows.

This function is one-to-one since

Therefore $|A| \leq |P(A)|$.

Now prove there is no bijection between A and $P(A)$ by contradiction.

This means there is no element in A that g will map to the set D .

So g is not onto all of $P(A)$. So g cannot be a bijection from A to $P(A)$

Therefore $|A| \neq |P(A)|$ meaning $|A|$ is strictly less than $|P(A)|$.

Example: Let A be the set of natural numbers and we define a function g mapping $\mathbf{N}$ to $P(\mathbf{N})$ as follows.